Zhifei Li

Ph.D. Student, Princeton CS

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Research Interests

My research is on **machine learning systems**, centered on algorithm–system **co-design**. I bring ML-systems methods to the most fundamental problems in coding tasks and beyond, treating agents as **first-class citizens** throughout. I currently work on the **environments** RL training depends on: tasks that evolve with the model, and high-fidelity simulation fast enough to generate robot data at scale.

Education

Princeton University

Ph.D. in Computer Science, advised by Prof. Tri Dao and Prof. Ravi Netravali — September 2026 - Present

Renmin University of China

B.S. in Computer Science, **Turing Honors Class** · GPA 3.8/4.0 (Top 5%) — September 2022 - June 2026

Ongoing Research

Tasks That Evolve with the Model: RL for Terminal Agents

RL post-training of terminal agents — August 2026 - Present

Human-written training tasks are expensive to build, yet they stop teaching once the model solves them. I built an online RL loop in which the training tasks evolve with the model, rewritten harder or easier from its own rollouts. RL on a curated task mix lifted a 9B model on Terminal-Bench 2 from 17% to 30%.

High-Fidelity Simulation Fast Enough for Robot Learning

Co-design of numerical methods and GPU kernels for tactile simulation — September 2026 - Present

Only high-fidelity simulation yields robot data that transfers to the real world, yet accurate contact and tactile simulators run slower than real time. By co-designing the numerical method with its GPU implementation, my general solver, with no per-scene tuning, runs 22–30× faster than the existing one at equal physical error (up to 66× on some scenes).

Research Experience

ByteDance Seed

Research Intern, Kernel Agent Training — June 2026 - August 2026

Sky Computing Lab, UC Berkeley

Research Intern, advised by Prof. Ion Stoica
worked with Prof. Joseph E. Gonzalez, Prof. Matei Zaharia — July 2025 - December 2025

- **SkyNomad** (**NSDI '27**, first author): multi-region spot scheduling, >50% cheaper than the prior SOTA
- **LEANN** (**MLSys '26 Best Paper**): vector index with 97% less storage; 13k+ GitHub stars
- **AI-driven systems**: led the Let the Barbarians In case study (+30%); co-built FrontierCS (**ICML '26**)

Publications

SkyNomad: On Using Multi-Region Spot Instances to Minimize AI Batch Job Cost

Zhifei Li*, *Tian Xia**, *Ziming Mao*, *Zihan Zhou*, *et al.*, *Yifan Qiao*, *Ion Stoica*
NSDI '27; [PDF]

LEANN: A Low-Storage Vector Index for Personal Devices

Yichuan Wang, **Zhifei Li**, *Shu Liu*, *Yongji Wu*, *et al.*, *Ion Stoica*, *Sewon Min*, *Matei Zaharia*, *Joseph Gonzalez*
MLSys '26 (Best Paper Award); [PDF]

PixelRAG: Web Screenshots Beat Text for Retrieval-Augmented Generation

*Yichuan Wang**, **Zhifei Li***, *Zirui Wang*, *Paul Teiletche*, *Lesheng Jin*, *Matei Zaharia*, *Joseph E. Gonzalez*, *Sewon Min*
ICLR '27 (under review); ICML '26 EMM-QA Workshop; [Website]

Let the Barbarians In: How AI Can Accelerate Systems Performance Research

*Audrey Cheng**, *Shu Liu**, *Melissa Pan*, **Zhifei Li**, *Shubham Agarwal*, *Mert Cemri*, *et al.*, *Matei Zaharia*, *Ion Stoica*
arXiv preprint, 2025; [PDF]; 100+ citations; featured on the SIGOPS blog

FrontierCS: Evolving Challenges for Evolving Intelligence

*Qiuyang Mang**, *Wenhao Chai**, **Zhifei Li***, *Huanzhi Mao**, *et al.*, *Alvin Cheung*
ICML '26; [PDF]; [Website]; used in the ByteDance Seed 2.1 model card

EvoX: Meta-Evolution for Automated Discovery

Shu Liu, *Shubham Agarwal*, *Monishwaran Maheswaran*, *Mert Cemri*, **Zhifei Li**, *et al.*, *Matei Zaharia*, *Ion Stoica*
COLM '26; [PDF]; 78 citations; 700+ GitHub stars

Open-Source Projects

LEANN: the Smallest Vector Index in the World

LEANN 13.0k⭐

Unified memory for personal AI on your own device, with 97% less storage — September 2024 - Present

- Led the open-source release: 40k+ downloads, 20 active external contributors, 600k+ views of its launch

PixelRAG: Web Screenshots Beat Text for RAG

PixelRAG 10.3k⭐

Reframes how AI reads the web: in its native form, pixels, not parsed text — January 2026 - Present

- Led the project as co-first author and shipped it as a hosted search API and a coding-agent skill

SkyPilot: Run AI on Any Infra

SkyPilot 10.7k⭐

Framework for running ML/AI workloads across any cloud infrastructure — September 2024 - Present

- Top 10 contributor (50+ merged PRs); built the High Availability Controller for SkyServe, adopted by Hypermode, IP Copilot and Liner

Service and Leadership

Reviewer and Artifact Evaluation Committee

IEEE TPAMI; ICML '26 EMM-QA; AEC: OSDI '25, ATC '25, '26, MLSys '26 — 2025 - Present

Introduction to Computer Systems (ICS)

Head Teaching Assistant — Fall 2024, Spring 2025

- Led 6 TAs in teaching systems curriculum that covered cache hierarchies and memory optimization to 200+ students

Renmin University Computer Association

President — July 2024 - July 2025

- Led 100+ members across 6 departments; organized 10+ tech talks reaching 5,000+ attendees

Cheese: Learning by Doing in the Agent Era

okcheese.com

Co-founder — September 2023 - Present

- Agents now do the entry-level work through which juniors grow into seniors; co-founded Cheese to rebuild that path in universities through project-based learning with AI
- Built a platform where students do real projects with teammates and agents, designed so people keep understanding what the agents did and stay the ones who judge; several thousand users across several institutions

Honors and Awards

Beijing Municipal Outstanding Graduate (top 5% at the provincial level) — June 2026

***Best Research Paper Award**, MLSys 2026* (for LEANN) — May 2026

Wu Yuzhang Scholarship (Renmin University's highest student honor, <10 recipients/year) — May 2026

Pioneer Medal, Renmin University (academic innovation leader; 8 awarded of 106) — May 2026

***First Place**, National System Capability Competition* (among **10,000** from 1,200 schools) — November 2025

Elite Collegiate Award, China Computer Federation (<100 recipients **nationally**) — August 2025

Dean's Scholarship, Gaoling School of AI (15 recipients out of 2,000) — May 2025

National Scholarship (Top 0.2% nationally) — September 2024

First-Class Scholarship for Social Service (48 recipients out of **35,000**) — September 2023

First Prize, National Olympiad in Informatics, Beijing (300 winners) — December 2019